

HOWARD PARK

Unity Developer | Network Engineer | AWS & Unity
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Hello! I'm a Unity developer of 7 years with strong knowledge of AWS and C#.

After Meta, I've decided to repursue my passion for game engineering and look for roles in server architecture or C-related game programming!
I have a history of constantly learning new languages and technologies and am excited for it every time.

WORK EXPERIENCE

Meta — Technical Artist

Decemeber 2023 - Decemeber 2024

- Wrote a Unity sandbox tool for sound engineers to test ambient and avatar sounds to optimize noise profiles in high and low traffic settings.
- Improved performance for Horizon Worlds (100k+ users) on Unity by writing shaders, optimizing in-house asset creation tools and profiling tools.
- Functioned cross-functionally with engineers, artists, and designers by scoping technical feasibility and requirements for new seasonal content.
- Performed asset conversions and visual fidelity tests between Horizon and Unity environments to improve asset creation pipelines.

FiveMics — Lead Unity Programmer

August 2022 - November 2023

- Worked alongside the CEO to design and implement the multiplayer vision for the FiveMics card game.
- Designed and implemented AWS architecture for the game's backend, using Unity Mirror to handle server-client networking.
- Developed secure REST APIs on API Gateway with Lambda functions to manage gameplay sessions and enforce server-sided rule handling.
- Organized user profiling with role-based access control policies and secret management using IAM and AWS Secrets Manager.
- Employed Amazon CLI to deploy Unity builds on CloudFormation for scalable infrastructure updates.
- Developed a full-stack user onboarding and registration flow with UX artists and designers using Cognito, DynamoDB, and Unity.
- Produced clean documentation for accessing AWS projects and CLI utilizations.

Ultraviolet — Senior Unity Developer

April 2021 - June 2022

- Led a small team of Unity developers to build a metaverse prototype for the Octi social media app.
- Collaborated with iOS engineers to integrate Unity builds with XCode and administer two-way communication using Unity SDK.
- Collaborated with ML engineers to implement analytics APIs to the game to monitor player activity and retention.
- Programmed two multiplayer games using Unity Mirror and AWS Lightsail to spin-up virtual private servers for testing.
- Employed interpolation techniques to synchronize animations, physics and client data across the server.
- Developed functional UI screens and prototypes for the UX team to test AR camera features, tap gestures, and min-games.
- Increased community engagement by 20%, reaching #1 in the social media category for two weeks.

Evava VR — Unity Developer (AR)

September 2020 - March 2021

- Produced AR experiences using ARKit and Unity for venues like South Coast Plaza and Fashion, including an AR Santa for the COVID holiday season and a bunny stomping game that was controlled with interactive floor projections.
- Built Unity prototypes demonstrating AR capabilities to shareholders and in business pitches to garner funding.
- Mentored colleagues and interns on proper Unity habits and tools.

Konami Digital Entertainment — Quality Assurance Tester

Dec 2019 - March 2020

- Reported and tracked bugs to developers with clear replication steps and conducted console TRC compliance tests for several titles including *Yu-gi-oh! Legacy of the Duelist: Link Evolution* and *Skelattack!*

Tandon Online — Unity Developer

June 2018 - August 2018

- Created an online educational game where students were challenged to build the tallest tower possible together using color-coded blocks that would self-destruct on contact with a same colored block. The project was developed with school faculty to teach organizational behavior principles and reward strategic teamwork with controlled chaos.
- Networked all server-client interactions using Unity HLAPI including player-specific blocks, crane mechanics and user-interface.
- Implemented interpolation strategies to accurately synchronize the positions and states of physics-based objects across clients to ensure smooth gameplay.
- Conducted weekly pitches on several design iterations and mechanics with faculty to reach project goals within reasonable scope.

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Education

New York University: Tandon School of Engineering (2015 - 2019)

Major: B.S. Integrated Digital Media

Minor: Game Engineering

Accolades: Dean's List 2016-2017

Relevant Courses: Game Programming, Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Ideation & Prototyping

Skills

Languages: C#, C++, Python, Javascript, Typescript, PostgreSQL, Java, OpenGL, HLSL

Frameworks / Tools: AWS, Supabase, Pandas, Numpy, Scikit-learn, Selenium, BeautifulSoup

Engines/Libraries: Unity, Unreal, Gamemaker, SFML C++, PyGame

Projects

LoL Draft Simulator — Supabase, Network Architecture, Data Management/Analysis, Full-stack Unity

November 2024 - Present

- Developed a collectable card game prototype on Unity where players can collect cards of real League of Legends competitors, build synergistic rosters, and compete in a series of strategic ban-pick tournaments to gain rewards.
- Processed user tokens using Supabase C# SDK on Unity to create a secure registration and login pipeline.
- Automated real-time data manipulation using typescript to improve data integrity and operational efficiency.
- Wrote Python scripts to programmatically scrape and import millions of player and game metrics using the CargoQuery API.
- Trained a logarithmic regression model using sklearn to calculate roster strength and apply categorical weighting for strategic game modes and gimmicks.

Technologies used: Supabase, Typescript, Javascript, PostgreSQL, Python, Unity/C#

AR Waifu — Augmented Reality, AI, Pathfinding

June 2023 - July 2023

- One month exploration project to learn AR, AI and pathfinding algorithms. The product is an augmented reality management-sim where players oversee AI-driven anime characters under a house by maintaining their relationships, health and household tasks.
- Created task queues and behavior trees for each character to automate a schedule of tasks and behaviors based on personality, agitation and relationships with other characters.
- Implemented a grid-based A* pathfinding algorithm to maneuver characters around the house.
- Added touch gestures to move and engage characters with interactable objects and characters.

Technologies used: Unity/C#, ARFoundations, iOS SDK

"HMDR VR" and "Grounded" — VR, Unity

June 2023 - July 2023

- Programmed puzzles, controls and gameplay mechanics on Unity for two narrative VR projects for game development classes.
- Designed a narrative system that tracked player progress and triggered changes in environment, sounds and level design for immersive gameplay.

Technologies used: Unity/C#, ARKit

Interests and Fun Facts

- I'm a competitive gamer. I've peaked top 500 in Marvel Rivals for 2 seasons, Masters in TFT and Grandmaster in Overwatch. Also, love League.
- I love working on cars. I've spent the last two years learning how to fix and maintain one while fully restoring my 2006 Acura RSX.
- I am learning to make and grind my own coffee.
- I am learning Brazilian Jiu Jitsu and getting rolled every other day.